

Feature Detail & Demo Reference

This document is the **single source of truth** for implementing the CS2 RCON web interface (this repo is a fork of [nokit](https://github.com/codevski/nokit) (<https://github.com/codevski/nokit>) / defuse). It captures everything observed in the public demo at <https://nokit.app/demo> (<https://nokit.app/demo>) so that developers **never need to visit the demo again**.

It is split into two parts:

1. **Feature Status Table** — a quick overview of what is actually implemented vs. what is currently only a static demo / planned.
2. **Detailed Feature Specifications** — an exhaustive, per-feature reference of every UI element, data field, column, interaction, chart, color code, status indicator, example value, and technical/implementation note recorded from the demo.

Demo versions observed: `nokit v0.1.0-rc.1` (footer), `relay v0.4.2` (sidebar footer). License MIT.
Deploy target: Docker image `ghcr.io/nokit/nokit:latest`, default web port `8080`, data volume `/data`, single ~14 MB Go binary running alongside `srcds` over loopback RCON.

Implementation will begin with the **Dashboard**.

Section 1 — Feature Status Table

Feature	Status	Notes
Login / Auth	✓ Implemented	Password-based login, session tokens in SQLite
Multi-server switcher	✓ Implemented	Add/remove servers (name, RCON host, password) via header
RCON Console	✓ Implemented	Full terminal UI: timestamps, color-coded output, ↑ ↓ history, Ctrl+L clear, Tab auto-complete (5000+ CS2 CVARs/commands), RCON macros sidebar with localStorage persistence, clickable history panel, live/paused scroll toggle, Copy session & Export
Dashboard	✓ Implemented	Live stat cards (CPU/tick/players) with sparklines, server status, quick actions, round info, recent output — polls <code>status / stats</code> over RCON every 6s
Live Logs	✗ Not built	Demo only / planned
Players	✗ Not built	Demo only / planned
Maps	✗ Not built	Demo only / planned
CVAR Presets	✗ Not built	Demo only / planned
Config Editor	✓ Implemented	File browser tree view, code editor with line numbers, unsaved changes tracking, save/reload/exec via RCON, support for .cfg and .json files
Plugins	✗ Not built	Demo only / planned
Scheduler	✗ Not built	Demo only / planned
Admin	✗ Not built	Demo only / planned

Section 2 — Detailed Feature Specifications

Global Layout & Chrome (present on all pages)

These elements are shared across every page and should be built as part of the app shell.

Top navigation bar (full width, dark)

Left cluster:

- **nokit** logo (green square “n” icon + wordmark) — top-left.
- **Server switcher dropdown**: label `fra1.mm`, subtext `5.39.42.118:27015`, with a small green online dot and a chevron (dropdown to switch servers / add-remove servers). See §11.
- **tickrate** indicator: `127.8`
- **players** indicator: `9/12`
- **map** indicator: `de_mirage`

Right cluster:

- **Search box**: placeholder `cvars, players...` with a `⌘K` / `K` keyboard-shortcut hint badge (command palette style search). See §13.
- Small **HK** / `⌘K` badge button.
- **External-link icon** (open server / launch).
- **Settings gear icon**.
- **User menu**: avatar + `m0use@` (current user), with dropdown.
- **Power / logout icon** (far right).

Left sidebar navigation (grouped)

- **SERVER**
 - Dashboard (home icon) — active/highlighted green
 - RCON Console — has a small `⌘K` / `HK` badge on the right
 - Live Logs
 - Players
- **CONFIGURATION**
 - Maps
 - CVAR Presets
 - Config Editor
- **SYSTEM**
 - Plugins — has an amber/orange numeric badge `1` (e.g. update/notification count)
 - Scheduler
 - Admin

Sidebar footer: `relay v0.4.2`, a throughput indicator `↑ 2.1 kb/s · ↓ 18 kb/s`, and a green `● WS` (WebSocket connected) status.

Page footer (bottom, full width)

- Left: `nokit v0.1.0-rc.1 · MIT · ★ github · docs · changelog`
 - Right (status strip): `● srcds OK · ● relay OK · 10/07/2026 · 16:16:14 · Europe/Berlin (UTC+2)` — shows service health, date, live clock, timezone. The footer clock ticks live (observed 16:16 → 16:30 across the session).
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1. Dashboard

Page title: **Dashboard**. Subtitle: fra1.mm · 5.39.42.118:27015 · connected via rcon .

Top-right of page: green ● **live** pill + ↻ **Refresh** button.

Top stat cards (4 across)

Card	Big value	Sub-line
PLAYERS	9 / 12 (green +2 delta badge)	last 5m
TICK RATE	127.8 hz	target 128
CPU	22 %	4-core · 0.88 load avg
RAM	2.1 / 8 GB	resident

Current round panel (left, ~2/3 width)

Header: **Current round** — right side: tick #88669 · uptime 11h 02m .

- Map thumbnail image with label de_mirage and a competitive badge overlay.
- Field rows (label → value):
- map → de_mirage (active pool)
- gamemode → competitive · MR12 + OT
- round → 8 / 24 – CT 4 : 3 T
- round time → 0:47 remaining
- warmup → ended · live
- session → 2h 14m · matchzy "live"

Quick actions panel (right, ~1/3 width)

Header: **Quick actions**. Buttons top-to-bottom:

- ↻ Restart server
- Segmented toggle: [match | practice | pug] — **match** selected (highlighted).
- 🔒 No password
- ✦ End warmup
- ⚡ Auto-shuffle teams
- ⏻ Stop server (red/danger styling)

Mini metric chart widgets (4 across, sparkline/area charts)

Widget	Range	Value	Chart color	Footer
Players	24h	9 / 12	green line	peak 12 · avg 6.4
CPU	5m	22%	blue area	peak 41% · sustained 28%
Memory	5m	2.1 GB	purple line	26% of 8 GB
Tick	5m	127.8 hz	orange line	target 128 · jitter 0.4

stats // rcon dump panel (bottom-left)

Header:  **stats // rcon dump** — right: refreshed 4s ago .

Tabular columns: CPU | NetIn | NetOut | Uptime | Maps | FPS | Players | Svms | +-ms | ~tick

Row values: 22.0 | 18.4 | 61.7 | 11h02m | 14 | 127.8 | 9 | 7.42 | 0.40 | 127.8

Below the table (key/value lines):

- version: 1.40.6.3 (cs2)
- os: linux x86_64 (debian 12)
- gotv: 1x spec on :27020
- protocol: v15 / build 9462
- vac: secure
- workshop: 16 maps · 281 MB

Recent events panel (bottom-right)

Header:  **Recent events** — right: view all → link.

Event rows (timestamp + actor + action):

- 12:51 m0use@ ended warmup
- 12:49 jaeger kicked h4rdy_woods — afk
- 12:40 m0use@ changed map: de_mirage
- 12:38 jaeger applied preset: MR12 (24+0T)
- 12:35 m0use@ edited gamemode_competitive.cfg
- 12:18 system GotvFix: bind failure

Implementation notes: dashboard/stats auto-refresh every ~4s. The stat cards and mini charts are driven by the `status` RCON dump plus a metrics history buffer. The Recent events panel mirrors the Admin → Audit log.

✓ **Current implementation status (this fork):** The Dashboard is built and is the default view when a server is selected (`web/src/pages/dashboard-page.tsx`), with a Dashboard/Console tab bar in `server-page.tsx` . It polls the `status` and `stats` RCON commands every 6s via the `useServerStats` hook (`web/src/hooks/useServerStats.ts`), parsing them with `web/src/lib/rcon-parse.ts` (tolerant of multiple CS2 output formats).

- **Stat cards** (`StatsCard` + `Sparkline`): Players (green), Tick rate (orange, target 128), CPU (blue area chart, with FPS sub-line), and RAM. Sparklines are rendered from a rolling in-memory history. RAM is shown as `n/a` because CS2's `stats / status` do not report server memory over RCON.
- **Server status** (`ServerStatus`): online/offline dot, name, address, map, hostname, players, uptime, version, OS — all parsed from `status / stats` .
- **Quick actions** (`QuickActions`): Restart server (`_restart` , confirmed), match/practice/pug segmented toggle (`exec <mode>`), clear password (`sv_password ""`), End warmup (`mp_warmup_end`), Auto-shuffle teams (`mp_scrambleteams 1`), and Stop server (`quit` , confirmed) — each sends the command over the existing `/api/servers/:id/rcon` endpoint and surfaces a success/error line.
- **Round info** (`RoundInfo`): shows map, phase (live/offline) and player count from `status` ; round number/score/timer are shown as `-` with a note that they will come from the Live Logs stream (not exposed by `status / stats`).
- **Recent output** (`RecentOutput`): shows the last lines of the most recent `status` poll and a link that switches to the full Console tab.

2. RCON Console

Page title: **RCON Console**. Subtitle: `tab to autocomplete · ↑↓ history · ctrl-l clears` .

Top-right controls: `[ live | paused ]` segmented toggle (live selected), `Copy session` button, `Export` button.

Console output area (left, large monospace terminal, ~2/3 width)

Green/amber terminal text. Commands echoed with timestamp prefix `rcon HH:MM:SS >` . Command values are syntax-highlighted (numbers/limits in color). Example session content:

```
- rcon 12:50:00 > status
- hostname: nokit | fra1.mm · map: de_mirage · players: 10/12 · pubkey: 8f12...ed94
- rcon 12:52:02 > mp_warmuptime 15
- "mp_warmuptime" = "15" ( def. "30" min. 0.000000 max. 90.000000 ) replicated - Time after a
new round...
- rcon 12:54:04 > sv_password
- "sv_password" = "*****" archive notify - Password to join the server.
- rcon 12:56:06 > mp_warmup_end
- [matchzy] warmup ended. live in 5...
- - - live - -
```

Input bar (bottom of console)

Prompt: `rcon >` monospace input field, pre-filled demo text `mp_war` . Has a small submit/enter afford-
ance on the right.

Quirk: in the demo the input is static — typing appends but no live autocomplete dropdown fires. The subtitle advertises tab-to-autocomplete, ↑ ↓ history, and ctrl-I to clear as intended behaviors.

RCON macros panel (right sidebar)

Header: ⚡ **RCON macros**. Each macro is a button showing a friendly label + the underlying command (greyed monospace):

```
| Label | Command |
|-----|-----|
| warmup → live | mp_warmuptime 15 ; m... |
| reset score | mp_restartgame 1 |
| knife → side pick | matchzy_knife |
| pause match | matchzy_pause |
| kick all bots | bot_kick |
| demo: start | tv_record demos/manu... |
| demo: stop | tv_stoprecord |
| fix stuck round | mp_unpause_match ; m... |
- + New macro... button at bottom (create custom macro).
```

History panel (right sidebar, below macros)

Header: 📜 **History**. Clickable recent commands list (re-run on click):

```
- status
- mp_warmuptime 15
- sv_password
- mp_warmup_end
- changellevel de_inferno
- tv_record demos/2026-05-21
```

✓ **Current implementation status (this fork):** The RCON Console is fully built as a real terminal UI. The component lives in `web/src/components/console.tsx` (exported as `Console`) and is rendered from the Console tab in `web/src/pages/server-page.tsx`. It uses the existing `POST /api/servers/:id/rcon` endpoint — no new backend routes were added.

- **Layout:** two-column split — a left terminal area (~65% width) and a right sidebar (~35% width).
- **Terminal output:** every command is echoed with a timestamp prefix `rcon HH:MM:SS >` and output is color-coded — green for the command prompt/echo, amber for CVAR values, red for errors, and white/muted for regular output.
- **Input bar:** monospace prompt `rcon >` with shell-style keyboard shortcuts — ↑ ↓ navigate command history, Ctrl+L clears the terminal, and Tab triggers autocomplete with a dropdown covering **5000+ CS2 CVARs/commands** (loaded from `web/src/data/cs2-commands.json`, sourced from [ArminC-CS2-Cvars](https://github.com/armync/ArminC-CS2-Cvars) (<https://github.com/armync/ArminC-CS2-Cvars>)). The autocomplete is capped at 50 suggestions per prefix for performance.
- **Live/paused scroll toggle:** in live mode the view auto-snaps to the newest line; paused mode lets the user scroll freely through the backlog without being pulled back to the bottom.
- **Session tools:** 📄 Copy session copies the full session log to the clipboard, and ⬇ Export downloads it as a `.txt` file.
- **RCON macros panel:** 8 built-in macros (warmup → live, reset score, knife round, pause match, kick bots, demo start, demo stop, fix stuck round) plus custom macro creation via the + New macro form. All custom macros are persisted in `localStorage` under the key `nokit_console_macros`.
- **History panel:** shows the last 20 unique commands (most recent first); clicking an entry pastes it into the input. History is persisted across reloads in `localStorage` under the key `nokit_console_history`.

3. Live Logs

Page title: **Live Logs**. Subtitle: srcds_logaddress · SSE to nokit-relay · 19 retained · 19 shown (shown count updates when filtering).

Top-right controls: green ● **LIVE** pill, || **Pause** button, ↓ **Download .log** button.

Filter tabs (with count badges)

All 19 · Chat 3 · Kills 4 · Rounds 5 · Connect / team 2 · Plugin 1

- **Functional:** clicking a tab filters the stream by event category and updates the “N shown” count (verified — Chat tab showed only the 3 chat lines).

Search / grep bar

- Input with placeholder: `grep` – e.g. `STEAM_1:0, "awp", defused` and a `/` regex toggle affordance at the right.
- Toggle checkbox: **hide rcon noise**.
- Far right: `tail -f` · 1.4k lines/min throughput label.

Log stream (monospace, color-coded by event type)

Each line: HH:MM:SS <TYPE> message . The event-type keyword is colored. Types seen: rcon , world , round , kill , chat , conn , team , srv , plug .

Example lines:

```
- 12:51:08 rcon > mp_warmup_end
- 12:51:08 world World triggered "Warmup_End"
- 12:51:09 round World triggered "Round_Start"
-
12:51:14 kill "m0use<2><STEAM_1:0:48211904><CT>" killed "zywoo<7><STEAM_1:1:11923844><T>"
with "ak47" (headshot)
- 12:51:23 chat "m0use<2><STEAM_1:0:48211904><CT>" say "gg wp"
- 12:51:25 kill "NEK0<5><STEAM_1:1:78231004><T>" killed "ferr<3><STEAM_1:1:91134472><CT>"
with "awp"
- 12:51:27 round Team "T" triggered "SFUI_Notice_Target_Saved" (CT "1") (T "0")
- 12:51:27 round World triggered "Round_End"
- 12:51:32 conn "jL<9><STEAM_1:0:90120431>>" connected, address "78.34.221.41:27015"
- 12:51:34 team "jL<9><STEAM_1:0:90120431><Unassigned>" switched from team <Unassigned> to
<TERRORIST>
- 12:51:40 chat "NEK0<5><STEAM_1:1:78231004><T>" say "rotate B"
- 12:51:55 rcon > status
- 12:51:55 srv hostname: nokit | fra1.mm · map: de_mirage · players: 10/12
- 12:52:01 kill "twist<6><STEAM_1:1:12349803><CT>" killed "bymas<8><STEAM_1:1:43712001><T>"
with "deagle" (headshot)
- 12:52:08 chat "olof<4><STEAM_1:0:55102941><CT>" say_team "info one toxic at long"
-
12:52:12 plug [WeaponPaints] failed to fetch skin metadata for STEAM_1:1:43712001 (timeout
5s)
-
12:52:30 kill "zywoo<7><STEAM_1:1:11923844><T>" killed "m0use<2><STEAM_1:0:48211904><CT>"
with "awp" (penetrated)
- 12:52:55 round Team "CT" triggered "SFUI_Notice_Bomb_Defused" (CT "1") (T "1")
- 12:53:01 rcon > changelevel de_inferno
```


Stream footer

tail) awaiting next event · backpressure 0 · last frame 1.2s ago with a green ● **SSE** connection indicator (right).

Implementation notes: logs are ingested via `srcds_logaddress` and streamed to the browser over **Server-Sent Events (SSE)** through nokit-relay. The stream shows backpressure and last-frame timing.

4. Players

Page title: **Players**. Subtitle: auto-refresh every 4s · 10 online · 4 on watchlist · 4 bans .
Top-right buttons: ⌵ Auto-shuffle , ⏻ Mass kick... , green + Add to watchlist .

Tabs (with count badges)

Live 10 · Watchlist 4 · Bans 4 · Session history

Common controls

- Search input: placeholder `search name, steamid, country, notes` .
- Team filter segmented toggle: [all | CT | T | spec] (all selected).
- 🗖 Filter button (right).

Live tab — table columns

NAME | STEAMID | TEAM | PING | TIME | K/D | COUNTRY

- **NAME**: some rows have an amber flag icon + a green `watch` badge (on watchlist). Rows with high ping shown in a warning color.
- **TEAM**: colored pill — CT (blue) / T (amber/orange).
- **PING**: e.g. `18ms` (high ping like `86ms` highlighted).
- **TIME**: connected time e.g. `1:24:11` .
- **K/D**: ratio e.g. `1.31` .
- **COUNTRY**: flag icon + 2-letter code.

NAME	STEAMID	TEAM	PING	TIME	K/D	COUNTRY
m0use_ (watch)	STEAM_1:0 :48211904	CT	18ms	1:24:11	1.31	SE
ferr	STEAM_1:1 :91134472	CT	22ms	1:24:08	0.94	DE
kqly.exe	STEAM_1:0 :32220110	CT	41ms	1:18:54	1.07	PL
twist	STEAM_1:1 :12349803	CT	24ms	1:17:02	1.18	SE
olof	STEAM_1:0 :55102941	CT	28ms	1:11:33	0.88	SE
NEKO	STEAM_1:1 :78231004	T	19ms	1:24:11	1.42	JP
zywoo*	STEAM_1:0 :11923844	T	26ms	1:23:01	1.95	FR
bymas	STEAM_1:1 :43712001	T	33ms	0:58:21	1.04	LT
jL	STEAM_1:0 :90120431	T	29ms	0:42:11	1.08	FI
h4rdy_woo ds (watch)	STEAM_1:1 :77124930	T	86ms	0:11:54	0.73	BR

Row hover/click reveals inline action buttons (right side of row): `profile` , `spectate` , `kick` , `ban...` (red/danger).

Watchlist tab — table columns

`NAME` | `STEAMID` | `REASON` | `STATUS`
`NAME`	`STEAMID`	`REASON`	`STATUS`
m0use_	STEAM_1:0:48211904	wallhack reports x4	● online · this server (green)
h4rdy_woods	STEAM_1:1:77124930	griefing — voted twice	● online · this server (green)
silent.knife	STEAM_1:1:20300912	team-damage spammer	● offline (grey)
badboy_3000	STEAM_1:0:65332100	racism — 1mo warn	● offline (grey)
- Each row has an amber flag icon. STATUS is a colored pill (green online / grey offline).

Bans tab — table columns

`NAME` | `STEAMID` | `REASON` | `BY` | `EXPIRES`
| `NAME` | `STEAMID` | `REASON` | `BY` | `EXPIRES` |
|-----|-----|-----|-----|
| lobotomy | STEAM_1:0:55512090 | wallhack — demo evidence | m0use@ | 2026-06-12 18:00 (amber)

|
rage_quitter	STEAM_1:1:11203044	griefing	admin	2026-05-24 04:00 (amber)
idiot	STEAM_1:0:99687700	racism — auto-detect	system	permanent (red)
spinbot	STEAM_1:1:00091123	aim hack — confirmed demo	m0use@	permanent (red)
 - EXPIRES: temporary bans show a date pill (amber); permanent bans show a red permanent pill.

Session history tab

Empty state message: No session history pruned yet. Last 30d retained.

Implementation notes: player list auto-refreshes every ~4s (parsed from status). Watchlist and bans are persisted server-side.

5. Maps

Page title: **Maps**. Subtitle: 12 standard · 4 workshop · current: de_mirage .

Top-right buttons: Sync workshop , Browse collection .

FAVORITES row

Section label FAVORITES with a ☆ show favs only toggle (right).

Compact favorite cards (thumbnail + name + mode badge):

- Mirage (active badge) · comp
- Inferno · comp
- Dust II · comp
- Overpass · comp
- aim_botz · practice

STANDARD POOL grid (4-across cards)

Each card: map thumbnail (radar-style with A/B bomb site markers), map name, a ☆ favorite star, and a mode badge (comp , hostage , practice). The active map has a green ● active badge and highlighted green border.

Maps: de_mirage (active), de_inferno , de_dust2 , de_nuke , de_overpass , de_anubis , de_vertigo , de_ancient , cs_office (hostage), cs_italy (hostage), de_train (comp), aim_botz (practice).

Map change interaction (verified): clicking a map card selects it as active — the green ● active badge and highlight move to the clicked card (both its pool card and favorites card update). This is how you change the current map.

WORKSHOP section

Label: WORKSHOP 281 MB CACHED · /HOME/CS2/SERVER/CSGO/MAPS/WORKSHOP .

- Workshop ID input: placeholder workshop ID — e.g. 3070900859 .
- host_workshop_map button.
- Green Download & switch button.

Workshop map cards (thumbnail + name + author + subscriber count + workshop ID + ☆):

Name	Author	Subs	Workshop ID
de_cache_redux	by shawn	281k subs	#3070900859
awp_lego	by 3kiksphilip	92k subs	#3070192884
surf_kitsune_ksf	by kitsune	14k subs	#3071182234
de_season	by sumamon	441k subs	#3072201991

MAP CYCLE editor (bottom)

Label `MAP CYCLE`. Ordered chip list (each chip has a numeric order + name + `x` remove):

`order: 01 de_mirage x · 02 de_inferno x · 03 de_nuke x · 04 de_dust2 x · 05 de_anubis x · 06 de_overpass x · 07 de_ancient x`

- `+ add` button to append a map.

- Right: `save mapcycle.txt` button (writes the cycle to mapcycle.txt).

6. CVAR Presets

Page title: **CVAR Presets**. Subtitle: `7 applied · changes preview as inline rcon · server.cfg untouched until "Persist"`.

Top-right buttons: `+ New custom preset`, `↓ Export .cfg`, green  `Apply all selected`.

Layout

Left ~2/3: preset categories, each a bordered group with a header `CATEGORY N / M APPLIED` and a row of toggle chips (checkbox pill; selected = green fill + checkmark). Each group ends with a `+ custom...` chip to add a custom preset.

Right ~1/3: **Inline preview** panel + **Quick switch** panel.

Preset categories & chips

- **ROUND SETTINGS 2 / 6 APPLIED:** `MR12 (24+0T) ✓ · MR15 (30) · Short (B09) · Warmup 15s ✓ · Warmup 60s · Freeze 10s · + custom...`
- **ECONOMY 2 / 4 APPLIED:** `Comp default ✓ · Pistol only · Rich start · 1:55 round ✓ · + custom...`
- **MOVEMENT 1 / 3 APPLIED:** `Default ✓ · Surf · KZ · + custom...`
- **SERVER 2 / 4 APPLIED:** `128 tick ✓ · 64 tick · LAN · Pure 2 ✓ · + custom...`
- **PRACTICE 0 / 4 APPLIED:** `Nade practice · Infinite nades · No-flash bots · Auto-respawn · + custom...`

Inline preview panel (right)

Header: `>> Inline preview` — right: `7 commands` (count updates live). Monospace numbered listing grouped by preset comment header:

- `#01 # MR12 (24+0T) → mp_maxrounds 24 ; mp_overtime_enable 1`
- `#02 # Warmup 15s → mp_warmuptime 15`
- `#03 # Comp default → mp_startmoney 800 ; mp_maxmoney 16000`
- `#04 # 1:55 round → mp_roundtime 1.92`
- `#05 # Default → sv_accelerate 5.5 ; sv_airaccelerate 12`
- `#06 # 128 tick → sv_minrate 128000 ; sv_maxupdaterate 128`

Footer: `✓ = to apply & persist` + green **Apply & persist** button.

Interaction (verified): toggling a preset chip live-updates the inline preview — enabling `MR15 (30)` inserted a `#02 # MR15 (30) → mp_maxrounds 30 ; mp_overtime_enable 0` block and the command counter incremented `7 → 8`.

Quick switch panel (right, below preview)

Header: `↔ Quick switch`. Full scenario presets (each with a keyboard shortcut badge):

- `Match — 5v5 MR12 + 0T` — green `active` badge
- `Practice — solo nades` — `⌘1`

- Pug — public retreat — %2
- LAN — sv_lan 1, no auth — %3

Implementation notes: presets preview as inline RCON commands; `server.cfg` is untouched until the user hits **Persist**. Applying sends the commands over RCON; persisting writes them to the config.

7. Config Editor

Page title: **Config Editor**. Subtitle (breadcrumb path): `/home/cs2/server/csgo/cfg/gamemodes/gamemode_competitive_server.cfg`.

Top-right: amber `unsaved` badge, `⌂ Reload from disk` button, `>_ exec via rcon` button, green `✓` Save & apply button.

File browser (left panel, tree)

Header: `cfg /` with a `+` (new file) button. Tree entries show a “last modified” age on the right:

- `autoexec.cfg` — 2d
- `server.cfg` — 1h
- `banned_users.cfg` — 3h
- `banned_ip.cfg` — 7d
- **gamemodes** (folder, 4)
 - `gamemode_competitive.cfg` — 12m
 - `gamemode_competitive_server.cfg` — (open/selected)
 - `gamemode_casual.cfg`
 - `gamemode_deathmatch.cfg` — 14d
- **matchzy** (folder, 3)
 - `mr12_competitive.cfg` — 8h
 - `live.cfg` — 0h
 - `warmup.cfg` — 0h
- **cs_sharp** (folder, 2)
 - `WeaponPaints.json` — 5d
 - `MatchZy.json` — 8h

Editor (right panel — code editor with line numbers)

Top status strip (right): `utf-8 · LF · 80 lines · last edit 12m ago by m0use@`.

Syntax-highlighted `.cfg` content with line numbers. Comments in green (`// ...`), cvar names one color, values another. Sample content:

```

1 // gamemode_competitive_server.cfg
2 // matchzy + tournament defaults - fra1.mm
3 // last edit: m0use@ - 12:35
4
5 mp_maxrounds          24
6 mp_overtime_enable    1
7 mp_overtime_maxrounds 6
8 mp_overtime_startmoney 16000
9 mp_freezetime         15
10 mp_roundtime          1.92
11 mp_warmuptime         15
12 mp_startmoney         800
13 mp_maxmoney           16000
14 mp_buy_anywhere       0
15 mp_team_intro_period  0
16
17 sv_pure               2
18 sv_minrate            128000
19 sv_maxupdaterate      128
20 sv_minupdaterate      128
21 sv_cheats             0
22
23 // matchzy
24 matchzy_minimum_ready_required 5
25 matchzy_kick_when_no_match_loaded 0
26 matchzy_demo_path      "matchzy_demos"
27 matchzy_chat_prefix    "[>nokit<]"

```

Quirk: in the demo the editor content is static — selecting another file in the tree does not swap the shown buffer. In the real implementation, selecting a file loads its buffer; the `unsaved` badge indicates dirty state; `Save & apply` writes to disk and can `exec` via `rcon`.

✓ **Current implementation status (this fork):** The Config Editor is fully built as a functional page. The component lives in `web/src/pages/config-editor-page.tsx` (exported as `ConfigEditorPage`) and is rendered from the `/servers/:id/config` route in `App.tsx`.

- **File browser tree:** Left sidebar (256px) with hierarchical tree view of config files. Mock data includes 7 files + 3 folders (gamemodes, matchzy, cs_sharp) with expand/collapse functionality. Each file shows “last modified” timestamp.
- **Code editor:** Full-width textarea with line numbers, monospace font, dark theme (bg-zinc-950). Supports multi-line editing with proper line height (1.5rem).
- **Unsaved changes tracking:** Compares current content with original content; shows amber “unsaved” badge when dirty. “Reload from disk” button resets to original content.
- **File operations:**
 - **Save & apply:** Saves file to server filesystem (TODO: backend integration)
 - **Reload from disk:** Discards changes and reloads original content
 - **exec via rcon:** Executes the file via RCON `exec <filename>` command
 - **Header breadcrumb:** Shows full path `/home/cs2/server/csgo{selectedPath}`
 - **Footer status:** Displays encoding (utf-8), line endings (LF), line count, and last edit metadata
 - **File selection:** Click any file in tree to load its content (currently shows demo content for all files)
 - **TODO:** Real file loading from server, syntax highlighting for .cfg/.json, create new file functionality, backend integration for saving files to server filesystem

8. Plugins

Page title: **Plugins**. Subtitle:

8 installed · 1 update available · 1 error · 1 warning · CS# 1.0.305 + Metamod 2.0.0-12 .

Top-right buttons: Reload all , Update all (1) , green + Browse registry .

Tabs (count badges)

Installed 8 · Browse 24+ · Error log 1 (Error log badge is red).

Installed tab

Controls: filter input filter installed – name, author, tag ; segmented status filter

[all | enabled | disabled | has update] (all selected); Install from .dll button (right).

Row layout: letter-avatar icon · name · version · status badge · tag badge(s) · author @handle · up-dated <age> · category · install path /addons/cs_sharp/plugins/<x> · description line · then on the right: an **enable/disable toggle switch** + Edit config + View logs + Details + red trash/delete icon.

Status badges: green enabled , grey disabled , amber warn , red failed , blue core , amber update → x.y.z .

Installed plugins:

Plugin	Ver	Status	Author	Updated	Category	Path	Description
CounterStrikeSharp	v1.0.305	enabled, core	@roflmuffin	12 May	core	/addons/cs_sharp/plugins/css	C# scripting host for CS2 dedicated servers. Loads .NET 8 plugin assemblies and exposes the gameserver event bus, RCON, and cvar APIs to managed code.
Metamod:Source	v2.0.0-12	enabled, core	@AlliedModders	08 May	core	/addons/cs_sharp/plugins/mm	Source engine plugin loader. Required by CounterStrikeSharp.
MatchZy	v0.8.7	enabled, update → 0.8.8	@shobhit-pathak	3d ago	competitive	/addons/cs_sharp/plugins/matchzy	Pug, scrim, and tournament match management with knife rounds, tactical pauses, and per-round demos.
KitsuneLab.MovementApi	v1.4.0	enabled	@kitsunelab	11d ago	utilities	/addons/cs_sharp/plugins/mov	Movement hook API used by surf, kz, and bhop plugins. Provides tick-accurate per-player movement callbacks.
WeaponPaints	v4.0	warn	@Nereziel	5d ago	fun	/addons/cs_sharp/plugins/paints	Skin / glove / knife chooser. Persists selection per SteamID via MySQL.
CS2-SimpleAdmin	v1.6.1	enabled	@daffyy	1mo ago	admin	/addons/cs_sharp/plugins/sa	In-game admin commands — kick, ban, gag, mute, freeze, slay. Compatible with SourceMod admins.cfg.
CallAdmin	v0.6.2	disabled	@rumblefrog	2mo ago	admin	/addons/cs_sharp/plugins/cadm	In-game report → Discord webhook with player context attached.
GotvFix	v0.3.0	failed	@whisper	14d ago	utilities	/addons/cs_sharp/plugins/gotv	TCP relay shim for GOTV — works around Valve's GOTV2 broken multicast on Linux 6.x kernels.

- The **failed** GotvFix row shows an inline red error box: [ERR] bind(0.0.0.0:27020) → EADDRINUSE – gotv relay disabled .

Browse tab (plugin registry / marketplace)

Left rail:

- **CATEGORIES:** All 24 · Admin tools 4 · Competitive 4 · Fun 4 · Utilities 4 · Maps 3 · Gamemodes 5.
- **FILTERS** (checkboxes): Has screenshots · CS2 compatible (checked) · MIT / Apache · No DB required.
- **SOURCES:** Registry 327 (active) · Custom git 3 · Local .dll.

Top bar: search `search 327 plugins – name, author (with / regex); sort options [ trending | most installed | recently updated | name ]` (trending selected).

FEATURED cards (icon · name · author · version · badge `featured / installed` · description · `↓ downloads` · `<age>` · green `install` button):

- CS2-RetakesAllocator v2.1.0 @b3none · featured · “Full retakes implementation — bombbsite rotation, automatic bomb planter, weapon allocator with...” · `↓47.4k` · 3d ago · [install]
- MatchZy v0.8.7 @shobhit-pathak · installed · “Pug, scrim & tournament match management with knife round, pauses, demos.” · `↓92.1k` · 3d ago
- WeaponPaints v4.0 @Nereziel · installed · “Skin / glove / knife chooser with web profile editor. MySQL-backed.” · `↓88.0k` · 5d ago
- Surf Timer v1.0.4 @kitsunelab · featured · “Per-map records, leaderboards, and per-zone splits with a TAS-quality replay bot.” · `↓21.6k` · 4d ago · [install]

ALL PLUGINS · 24 (row: icon · name · version · author · `↓downloads` · updated · category · tag chips · `Install` button + external-link icon):

- AccessChecker v1.1.0 @j00nfist · `↓24.6k` · 5d ago · Admin tools · `#groups #auth` · “Steam group + flag-based access control. Verifies via Steam Web API on connect with 5-minute cache.”
- VoteManager v2.0.1 @riffrrri · `↓18.2k` · 2w ago · Admin tools · `#voting` · “Veto, map vote, mute vote, kick vote. Configurable thresholds and player-side menus.”
- SpectatorTools v0.9.4 @b3none · `↓9.7k` · 1mo ago · Admin tools · `#debug` · “Force spectate, freeze, no-clip, and god-mode admin tools.”

Error log tab

Controls: search `search plugin output – message, plugin name, level`; segmented level filter `[ all | errors | warnings | info ]`; `↓ Export` button (right).

Monospace log lines `YYYY-MM-DD HH:MM:SS [LEVEL] PluginName – message`. Levels color-coded: `[WARN]` amber, `[ERR]` red, `[INFO]` grey/blue.

-

`2026-05-21 12:48:02 [WARN] WeaponPaints – failed to fetch skin metadata for STEAM_1:1:43712001 (timeout 5s)`

- `2026-05-21 12:18:54 [ERR ] GotvFix – bind(0.0.0.0:27020) → EADDRINUSE – gotv relay disabled`

- `2026-05-21 11:30:11 [INFO] MatchZy – matchzy: live config "mr12_competitive.cfg" applied`

- `2026-05-21 11:08:31 [INFO] CounterStrikeSharp – plugin host started · 8 plugins loaded in 412ms`

- `2026-05-21 10:55:14 [INFO] MatchZy – autoloader: warmup.cfg → live.cfg transition configured`

- `2026-05-21 10:42:11 [INFO] WeaponPaints – mysql: connected to 127.0.0.1:3306 (db=weapons, latency 1.4ms)`

- `2026-05-21 10:39:02 [WARN] GotvFix – falling back to TCP relay – kernel multicast probe failed`

- `2026-05-21 10:38:54 [ERR ] GotvFix – unable to subscribe to GotvFix.RelayPort (27020) – retrying in 30s`

Footer: `plugin-log` `) last 24h · 1 err · 2 warn · 12 info · 8 lines` + `clear` button.

9. Scheduler

Page title: **Scheduler**. Subtitle: `4/5 active · system tz: Europe/Berlin · cron via nokit-relay`.

Top-right buttons: `⚡ Run now...`, green `+` `New task`.

Jobs table (left, main)

Columns: NAME | SCHEDULE | COMMAND | LAST | NEXT . Each row has a status dot on the left (green = active, grey = disabled). NEXT run is bold when upcoming.

```
| ● | NAME | SCHEDULE (cron) | COMMAND | LAST | NEXT |
|-----|-----|-----|-----|-----|
| green | Nightly restart | 0 5 * * * | sv_shutdown | 2026-05-21 05:00 | 2026-05-22 05:00 |
| green | Warmup reset | */30 * * * * | mp_warmup_start | 12:30 | 13:00 |
| green | Workshop sync | 0 4 * * 0 | ds_workshop_sync | 2026-05-18 04:00 | 2026-05-25 04:00 |
| green | Daily backup | 0 3 * * * | exec ./backups/snap.sh | 2026-05-21 03:00 | 2026-05-22 03:00 |
| grey | Pug auto-shuffle | 0 19 * * 5 | mz_shuffle | 2026-05-16 19:00 | 2026-05-23 19:00 (disabled) |
```

RECENT RUNS table (below jobs)

Columns: WHEN | TASK | DURATION | RESULT . RESULT is a colored pill: green ok , amber N retries , grey disabled .

```
| WHEN | TASK | DURATION | RESULT |
|-----|-----|-----|-----|
| 12:30:00 | Warmup reset | 0.08s | ok |
| 12:00:00 | Warmup reset | 0.07s | ok |
| 05:00:00 | Nightly restart | 14.2s | ok |
| 03:00:00 | Daily backup | 2m 04s | ok |
| 04:00 (Sun) | Workshop sync | 8m 41s | 3 retries (amber) |
| 19:00 (Fri) | Pug auto-shuffle | — | disabled (grey) |
```

Presets panel (right)

Header: ✦ **Presets**. Clickable preset cards (name + cron/trigger + command, monospace grey):

- Nightly restart — 0 5 * * * → sv_shutdown
- Warmup reset every 30m — */30 * * * * → mp_warmup_start
- Pre-match script — on-event match → matchzy_load mr12
- GOTV demo cleanup — 0 6 * * 0 → find demos -mtime +30 -delete
- Discord ping if down — on-event down → webhook discord_alerts

Next 24h panel (right, below presets)

Header: ⏴ **Next 24h**. Timeline list of upcoming runs (time + task):

- 13:00 Warmup reset
- 13:30 Warmup reset
- 14:00 Warmup reset
- — hourly until 04:30 — (collapsed range indicator)
- 05:00 Nightly restart
- 03:00 Daily backup

Implementation notes: cron scheduling runs via nokit-relay. Jobs support both cron expressions and on-event triggers (match, down). System timezone shown: Europe/Berlin.

10. Admin

Page title: **Admin**. Subtitle: 4 users · audit retention 90d · Discord webhook configured · GOTV port 27020 .

Top-right buttons: ↓ Export audit log , green + Invite admin .

Tabs

Users & roles 4 · Audit log 10 · Integrations · GOTV & demos .

Users & roles tab

Users table columns: USER | EMAIL | ROLE | MFA | LAST SEEN . ROLE is a colored pill; MFA is green on / amber off pill.

USER	EMAIL	ROLE	MFA	LAST SEEN
m0use@	m@m0use.dev	owner (green)	on	12:51 — now
jaeger	p.j@axum.io	admin (blue)	on	2h ago
kira	kira@scrim.gg	admin (blue)	off (amber)	1d ago
ref-bot	discord://ref	moderator (grey)	on	4d ago

Role matrix panel (right) — Header **Role matrix**. Columns: CAPABILITY | OWNER | ADMIN | MOD (✓ = granted, · = not):

CAPABILITY	OWNER	ADMIN	MOD
rcon console	✓	✓	·
kick / ban	✓	✓	✓
edit configs	✓	✓	·
install plugins	✓	✓	·
manage admins	✓	·	·
manage servers	✓	·	·

Audit log tab

Controls: search search by user, action, target, time ; time-range segmented filter

[today | 7d | 30d | all] (today selected); type filter [all | writes | signin] (writes selected).

Columns: WHEN | WHO | ACTION | TARGET + a status check (✓ ok / ⚠ warning icon at far right). ACTION is a colored keyword pill (cmd , kick , plugin , map , preset , config , ban , signin).

WHEN	WHO	ACTION	TARGET	
12:51:08	m0use@	cmd	mp_warmup_end	✓
12:49:51	jaeger	kick	h4rdy_woods (STEAM_1:1:77124930)	afk ✓
12:48:02	system	plugin	WeaponPaints warn: fetch timeout	⚠
12:40:11	m0use@	map	de_mirage (was de_inferno)	✓
12:38:00	jaeger	preset	apply: MR12 (24+OT)	✓
12:35:12	m0use@	config	edit gamemode_competitive.cfg (+4 -2)	✓
12:30:00	cron	cmd	mp_warmup_start	✓
12:18:54	system	plugin	GotvFix bind failure	⚠
12:14:33	kira	ban	lobotomy — wallhack — 21d	✓
12:02:11	m0use@	signin	from 84.124.221.9 — chrome/macOS	✓

Integrations tab (4 cards)

Discord webhook — status ● connected (green):

- url `https://discord.com/api/webhooks/123•••••/k83a1••••••••••`
- channel `#nokit-fra1`
- events `match start · match end · server down · plugin error · ban`
- last `12:51 – match started`
- Buttons: `test message`, `edit`, `red disconnect`.

Steam workshop API — status ● key valid (green):

- key `A1F8••••••••••••••••E0`
- quota `4,221 / 100,000 calls (24h)`
- collection `ws://3142099912 – 16 maps`
- last sync `2026-05-18 04:08 · 8m 41s · 0 errors`
- Buttons: `rotate key`, `green sync now`.

Steam game-server login token — status ● active (green):

- app `730 (cs2)`
- token `F38C1B••••••••••••••••`
- label `nokit-fra1`
- added `2026-04-11`

Backup target — status ● S3 :

- bucket `s3://nokit-backups`
- prefix `fra1.mm/`
- retention `30d (cfg) · 14d (demos)`
- last `03:00 · 412 MB · ok`

GOTV & demos tab

Demo files table columns: `FILE | SIZE | DURATION | DLS (downloads):`

`| FILE | SIZE | DURATION | DLS |`

`|-----|-----|-----|-----|`

`| 2026-05-21_de_mirage_mr12_team-0_vs_team-B.dem | 94.2 MB | 46m | 2 |`

`| 2026-05-20_de_inferno_scrim_axum_vs_nbk.dem | 102.8 MB | 52m | 0 |`

`| 2026-05-20_de_nuke_pug.dem | 78.4 MB | 38m | 1 |`

`| 2026-05-19_de_anubis_eu-quals.dem | 121.0 MB | 1h 04m | 8 |`

GOTV settings panel (right) — Header **GOTV settings** (cvar → value):

- `tv_enable 1`
- `tv_port 27020`
- `tv_maxclients 32`
- `tv_delay 90s`
- `tv_autorecord 1`
- `tv_chatgroup "nokit-gotv"`
- `status ● 1 spectator · 0.4 mbps (green)`
- **DEMO RETENTION:** `keep [ 14 days ] then [ upload to S3 ]` (editable pills).

11. Multi-server switcher

Opened from the top-left server dropdown (`fra1.mm · 5.39.42.118:27015`).

Dropdown panel:

- Section label **SERVERS**. Each row: online status dot + name + ip:port · region + player-count badge + × remove button.

| Name | Address | Region | Players |

|-----|-----|-----|-----|

| fra1.mm (active) | 5.39.42.118:27015 | Frankfurt | 9/12 |

| seal.scrim | 10.0.1.22:27015 | Seattle | 10/10 |

| home.lan | 192.168.1.40:27015 | Local | 3/10 |

| practice-aws | 54.218.7.92:27015 | us-west-2 | 0/8 |

| retake-old | 88.99.10.4:27015 | Hetzner | — (offline, grey dot) |

- Section label **ADD SERVER** — inline form: name text input, ip:port text input, green **Add** button, and a rcon password input below (full-width).

- Clicking a server row switches the active server (updates top-bar tickrate/players/map and all pages).

Corresponds to CLI `srv > add [hostname] [IP:PORT]` .

12. Login / Auth & Session

The public demo (`nokit.app/demo`) is a pre-authenticated single-page mock — there is **no separate /login route** (visiting `/login` returns 404). Auth is documented rather than shown as a live form:

- **No SaaS accounts / no telemetry**. Self-hosted only.

- Initial admin set via env var in docker-compose: `NOKIT_ADMIN: m0use@local` .

- Authentication is via **SteamID or email**, with **MFA support** (see Admin → Users & roles: MFA on / off per user).

- **Role-based access control**: Owner / Admin / Moderator, enforced by the per-capability role matrix (see Admin).

- Sign-in events are recorded in the Audit log (e.g. `signin` — from `84.124.221.9` — `chrome/macOS`).

- Current session identity is shown in the top bar as `m0use@` with an avatar; a **power/logout icon** sits at the far right of the top bar. (In the demo these are static — the user menu does not open a drop-down.)

- Session/connection state surfaced globally: sidebar footer ● WS indicator (note: docs describe live log transport as **SSE**, and RCON over loopback; the UI also shows a WS status pill) and footer ● srcds OK · ● relay OK .

This fork's implementation note: Auth is currently password-based login with **session tokens persisted in SQLite** (see Feature Status Table §1).

13. Global search / Command palette

- Top-bar search field placeholder `cvars, players...` with a ⌘K keyboard shortcut badge.
- Intended as a command palette to jump to cvars/players/pages. In the demo, activating it routes to the **RCON Console** (the ⌘K shortcut and the sidebar RCON ⌘K badge are linked). No floating overlay palette renders in the static demo.

Cross-cutting UI / Design notes (for re-implementation)

- **Theme:** dark UI (near-black background `#0d0d0f` -ish), green primary accent (buttons, active nav, online dots), monospace fonts for logs/console/cvars, sans-serif for chrome.
 - **Status color language:**
 - **green** = ok / online / enabled / active
 - **amber/orange** = warning / T-side / update-available / temporary
 - **red** = error / failed / ban / danger / permanent
 - **grey** = disabled / offline
 - **blue** = CT-side / admin-role / core-plugin / info
 - **Common layout pattern:** page header (title + descriptive subtitle on the left, action buttons on the right) → optional tab bar with count badges → main content (often a left main panel + right sidebar of secondary panels).
 - **Segmented toggles** used widely (match/practice/pug, all/CT/T/spec, live/paused, time ranges).
 - **Tables** use uppercase small-caps column headers, colored pill badges for status/role/team, and reveal inline row-action buttons on hover/selection.
 - **Every mutating action** is designed to be logged to the Audit log with actor + action + target.
 - **Refresh cadence:** dashboard/stats and player list auto-refresh every ~4s; logs stream via SSE.
 - **Update frequency example:** page footer clock ticks live (observed 16:16 → 16:30 across the session), timezone `Europe/Berlin (UTC+2)`.
 - **Version strings:** app `nokit v0.1.0-rc.1`, relay `v0.4.2`, CS# `1.0.305`, Metamod `2.0.0-12`, CS2 build `1.40.6.3` / protocol v15 build 9462.
 - **Deploy target:** Docker image `ghcr.io/nokit/nokit:latest`, default web port `8080`, data volume `/data`, runs alongside `srcds` (loopback RCON). Single ~14 MB Go binary.
-

This document was compiled from observations of the nokit demo. Keep it updated as features move from  (demo/planned) to  (implemented).